

Article

Deep Multimodal Learning for Heart Sound Classification Using CNN, Transformer, and BiLSTM with Attention

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Abstract

Phonocardiogram (PCG) signals offer a non-invasive, low-cost screening tool for cardiovascular diseases. However, their noisy and non-stationary nature makes automated classification challenging, and traditional methods often fail to capture complex spectral-temporal patterns. This study proposes a multimodal deep learning architecture for the binary classification of heart sounds (Healthy vs. Unhealthy). The hybrid model integrates Convolutional Neural Networks (CNNs), Transformer encoders, and Bidirectional Long Short-Term Memory (BiLSTM) networks with an attention mechanism. It utilizes an early-fusion feature extraction pipeline combining MFCCs, Mel-spectrograms, and Chroma descriptors. To ensure robust evaluation and prevent data leakage, SMOTE is applied exclusively to the training folds within a strict zero-leakage, patient-wise 5-fold cross-validation protocol. The proposed framework demonstrates exceptional performance, achieving an average accuracy of 91.67%, a sensitivity of 80.95%, a specificity of 94.46%, and an AUC-ROC of 96.50%. An ablation study confirms that integrating Transformer and BiLSTM modules significantly enhances diagnostic stability over baseline CNNs. Furthermore, with exactly 858,434 parameters (3.27 MB) and interpretable attention maps, this highly optimized model provides a robust assistive solution suitable for deployment in digital stethoscopes and mobile telemedicine systems.

Keywords: Phonocardiogram; SMOTE; CNN–Transformer–BiLSTM



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1. Introduction

Cardiovascular diseases (CVDs) remain the leading cause of mortality worldwide, placing a tremendous burden on public health systems [1,2]. Specifically, evidence from the Global Burden of Disease (GBD) project highlights heart valve defects (HVDs) as a significant driver of increased mortality, chronic disability, higher hospitalization rates, and escalating healthcare-related economic costs [1]. Early detection is therefore paramount to reducing mortality rates and preventing severe complications.

The human heart contains four major valves—aortic (AV), pulmonary (PV), mitral (MV), and tricuspid (TV)—whose primary role is to ensure unidirectional blood flow by preventing backflow [3]. The cyclic opening and closure of these valves, together with hemodynamic turbulence within the cardiac chambers and great vessels, generate characteristic heart sounds. These acoustic signals can be captured and analyzed as phonocardiograms (PCGs). In clinical practice, experienced practitioners rely on stethoscope

auscultation to detect variations in the timing, intensity, and pitch of heart sounds, enabling rapid screening for valvular abnormalities such as stenosis or regurgitation.

In this context, cardiac auscultation via the PCG serves as an essential first-line clinical screening tool. By recording the mechanical activity of the heart, PCG provides a non-invasive, cost-effective, and widely accessible method, particularly valuable in resource-constrained environments where access to advanced imaging modalities may be limited. Clinicians typically assess the two fundamental heart sounds, S1 and S2, as key markers of cardiac-cycle events. The first component, S1, is mainly associated with mitral and tricuspid valve closure, whereas the second component, S2, corresponds to aortic and pulmonary valve closure. A complete cardiac cycle extends from the onset of S1 to the onset of the next S1, with S1 generally exhibiting a longer duration and lower-frequency content than S2 [3,4].

During the intervals between S1 and S2, disturbed or turbulent blood flow may generate additional acoustic components that appear as abnormal patterns in PCG waveforms. These phenomena, commonly referred to as heart murmurs, are typically linked to valvular defects [4]. However, PCG recordings are inherently non-stationary and susceptible to environmental noise and inter-subject variability. Valvular pathologies such as aortic stenosis (AS), mitral stenosis (MS), mitral regurgitation (MR), and mitral valve prolapse (MVP) can produce highly similar PCG patterns, making robust automated classification a highly challenging task.

While numerous machine learning architectures have been proposed to automate this process, many existing studies suffer from methodological flaws, such as record-wise data splitting, which inadvertently causes data leakage and artificially inflates performance metrics. To address this, our study introduces a highly optimized, lightweight multimodal deep learning architecture combining Convolutional Neural Networks (CNNs), Transformers, and Bidirectional Long Short-Term Memory (BiLSTM) networks. The primary contributions of this work are threefold:

1. We establish a highly rigorous, zero-leakage patient-wise cross-validation pipeline, ensuring that all dataset augmentation (SMOTE) is strictly confined to the training folds.
2. We utilize a high-resolution early-fusion strategy of multimodal features to achieve robust binary classification (Healthy vs. Unhealthy) with a highly compact model suitable for edge-device deployment.
3. We conduct a comprehensive ablation study that justifies the hybrid architecture, demonstrating how the BiLSTM maximizes sensitivity while the Transformer maximizes specificity, culminating in a highly stable clinical AUC-ROC of 96.50%.
4. We provide visual interpretability of the model's decision-making process through attention maps, validating its clinical relevance.

The remainder of this paper is organized as follows: Section 2 reviews the related academic literature. Section 3 details the materials and methods, covering the dataset description, zero-leakage preprocessing pipeline, feature extraction techniques, and the proposed hybrid architecture. Section 4 presents the experimental results, including robust performance metrics and state-of-the-art comparisons. Section 5 discusses the interpretation of the findings, model efficiency, and clinical applicability. Finally, Section 6 concludes the paper and outlines potential directions for future research.

2. Related Work

However, the automated analysis of PCG signals presents substantial challenges. These recordings are inherently noisy, non-stationary, and exhibit high inter-subject variability, often depending heavily on acquisition conditions and hardware. To address these

complexities, research has historically relied on traditional Machine Learning (ML) approaches [3,5–8]. Algorithms such as Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), and Random Forests have been extensively explored. For instance, Goda and Hajas employed a Dual-Tree Complex Wavelet Transform to extract features fed into an SVM classifier, achieving an accuracy of 81.2% [9]. Similarly, other studies utilizing Mel-frequency cepstral coefficients (MFCC) combined with SVMs or simple Feed-forward Neural Networks (FNN) have reported accuracies generally ranging from 79% to 86% on complex datasets like PhysioNet 2016 [10,11].

While these conventional methods provided a foundational baseline, they are fundamentally limited by their reliance on handcrafted features. This manual extraction process requires specific domain expertise and often fails to fully capture the rich, high-dimensional spectral and temporal patterns inherent in heart sound signals.

The advent of Deep Learning (DL) [12] marked a paradigm shift by enabling automatic feature learning from raw data or spectrograms. Convolutional Neural Networks (CNNs) initially demonstrated great promise in extracting local invariant patterns. A notable study by Rubin et al. applied a 2D CNN to MFCC spectrograms, achieving an accuracy of approximately 84% during the PhysioNet/CinC 2016 challenge [13,14]. Subsequent research focused on optimizing these architectures for better performance and computational efficiency. For example, Xiao et al. and other researchers proposed a 1D CNN architecture that improved accuracy to 93% [15–17]. More recently, Zhu et al. (2024) introduced a lightweight CNN model (LEPCNet) tailored for wearable devices, reaching 93.1% [18,19]. Even more complex hybrid attempts, such as the PCTMF-Net by Wang et al. (2023), which integrates CNNs with Transformers, have achieved similar performance levels (~93%) [20].

Despite these significant advancements, ensuring consistent generalization on heterogeneous datasets remains a challenge. While some recent studies, such as Khan et al. [21] and Humayun et al. [22], have reported accuracies exceeding 95% using complex transfer learning pipelines [23], these approaches often require heavy computational resources or rely on fixed train-test splits that may not fully reflect real-world clinical variability. Purely convolutional models may miss the long-range temporal dependencies [24] necessary to understand the complete cardiac cycle, while standard hybrid models do not always effectively fuse spatial and temporal features. Consequently, there remains a critical need for advanced multimodal architectures capable of seamlessly integrating spatial, frequency, and temporal information while maintaining robust generalization capabilities.

In this study, we propose a novel Deep Multimodal Learning framework that synergistically combines the feature extraction power of 1D-CNNs, the sequential modeling capabilities of BiLSTM (Bidirectional Long Short-Term Memory) networks [25], and the global context awareness of Transformers equipped with an attention mechanism [26]. By leveraging an early fusion of multimodal acoustic features (MFCC, Mel-spectrogram, and Chroma), our approach is explicitly designed to capture not only the subtle local spectral characteristics of heart sounds but also their global temporal context. To ensure the utmost reliability of our results and strictly prevent data leakage—a prevalent bias in existing literature—our method is validated using a rigorous zero-leakage patient-wise 5-fold cross-validation strategy (GroupKFold). Tested on the benchmark PhysioNet/CinC 2016 dataset [10], our framework achieves a highly robust classification accuracy of 91.67% and an outstanding AUC-ROC of 0.9650. This result represents a superior clinical trade-off between high diagnostic precision and true model generalization compared to state-of-the-art approaches, highlighting its potential as a reliable tool for the computer-aided screening of cardiovascular conditions.

3. Materials and Methods

3.1. Dataset Description

This study utilizes the large-scale PhysioNet/Computing in Cardiology (CinC) Challenge 2016 dataset [10], a widely recognized benchmark for heart sound classification. The dataset reflects real-world clinical conditions with significant heterogeneity, including background noise, motion artifacts, and varying stethoscope placements (mitral, aortic, tricuspid, and pulmonary areas).

- **Composition and Class Imbalance:** Following initial loading and validation, the dataset utilized in this study consists of a total of 3241 PCG recordings. A significant natural class imbalance exists between healthy and pathological samples (Figure 1). To ensure a highly rigorous evaluation and entirely avoid the selection bias inherent to static train/validation splits, we discarded the fixed split approach. Instead, the entire cohort of 3241 samples is evaluated dynamically through our zero-leakage 5-fold patient-wise cross-validation strategy.
- **Recording Duration:** While the original raw recordings vary significantly in length (typically ranging from 5 to over 35 s), all signals were systematically standardized into 5-s segments to ensure consistent input tensor dimensions for the neural network, capturing enough cardiac cycles for robust feature extraction while minimizing computational overhead.
- **Sampling Rate:** Crucially, as per the original dataset specifications, the native sampling frequency of all PCG recordings is **exactly 2000 Hz** [10]. To maintain the absolute integrity of the acoustic features and avoid introducing any aliasing artifacts, no down-sampling or upsampling was performed. All audio signals were strictly processed at their native **2000 Hz** sampling rate throughout the entire experimental pipeline.

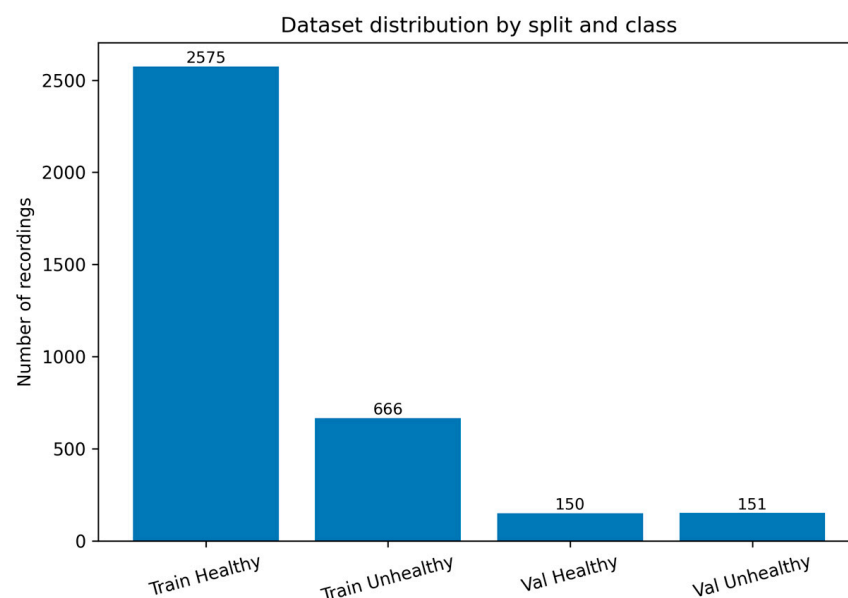


Figure 1. Distribution of the PhysioNet/CinC 2016 dataset classes showing the imbalance between Normal and Abnormal recordings.

3.2. Data Preprocessing

Given the inherently non-stationary nature of PCG signals, a robust preprocessing pipeline is critical to removing environmental noise while preserving the integrity of the S1 and S2 heart sounds (Figure 2). To ensure complete consistency and maintain high-fidelity acoustic features, the proposed pipeline consists of the following sequential stages:

- **Length Standardization:** To ensure uniform input dimensions for the neural network architecture, all recordings were systematically standardized to exactly 5 s. Signals shorter than 5 s were zero-padded, while longer signals were truncated. At the native 2000 Hz sampling rate, this yields a consistent vector of exactly 10,000 discrete time steps per audio segment.
- **Denoising and Filtering:** Background noise, sensor friction, and breathing sounds were suppressed using an advanced spectral gating algorithm. This method estimates the ambient noise profile directly from the signal and mathematically subtracts it from the active heart sound regions. This filtering approach effectively eliminates low-frequency artifacts without distorting the essential acoustic morphology required for the classification task.
- **Normalization:** To mitigate the extreme inter-subject amplitude variability caused by different stethoscope placements and body mass indexes, the denoised signals were normalized. Amplitude normalization was applied to scale the waveform boundaries, followed by mean subtraction to strictly center the signal at zero, thereby standardizing the amplitude range across the entire dataset.

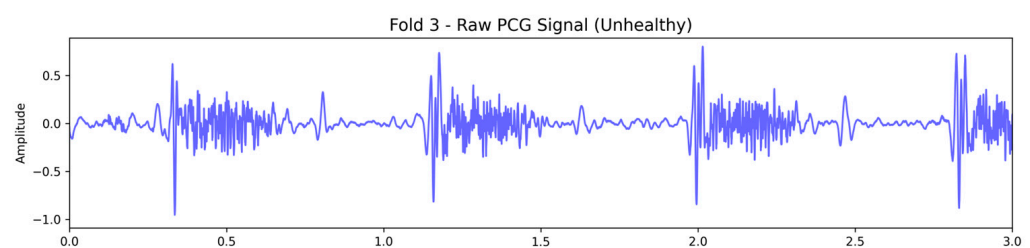


Figure 2. Example of a preprocessed Phonocardiogram (PCG) signal. While the model processes standardized 5-s recordings, the temporal window displayed here is restricted to a 3-s zoomed-in view to clearly visualize the distinct cardiac cycles, specifically the S1 and S2 heart sounds, without visual clutter. The x-axis represents Time (s) and the y-axis represents normalized Amplitude.

3.3. Class Balancing Using SMOTE

The PhysioNet/CinC 2016 dataset is highly imbalanced [10], with a disproportionate number of normal recordings compared to abnormal ones. To address this challenge and prevent the model from biasing toward the majority class, we employed the Synthetic Minority Over-sampling Technique (SMOTE) [27].

Crucially, to strictly prevent data leakage and ensure a robust evaluation, SMOTE was applied exclusively to the extracted feature space within the training set of each cross-validation fold. Since our model requires a 2D sequence input (timesteps $\times$ features), the SMOTE algorithm was implemented using a flatten-and-reshape strategy. Specifically, the training sequences (157×92) were first flattened into high-dimensional 1D vectors for the generation of synthetic minority samples. Once the classes were balanced, these vectors were reshaped back into their original 2D temporal structure. This approach ensures that the synthetic samples maintain the necessary structural alignment between timesteps and multimodal features for the hybrid architecture.

The validation folds were kept entirely pristine, unaugmented, and naturally imbalanced to accurately evaluate the model's true generalization performance in real-world clinical conditions.

3.4. Multimodal Feature Extraction

3.4.1. Mel-Spectrogram

The Mel-spectrogram represents the acoustic energy of the signal mapped onto the Mel scale, which mimics human auditory perception by utilizing a non-linear frequency

resolution [28]. The Mel scale m is mathematically related to the linear frequency f (in Hz) by the following equation:

$$m = 2595 \log_{10}\left(1 + \frac{f}{700}\right) \quad (1)$$

where f is the frequency in Hertz and m is the corresponding Mel frequency.

For each audio frame, the Short-Time Fourier Transform (STFT) is computed, and the power spectrum is mapped onto 128 Mel-frequency bins ($N_{mels} = 128$). The resulting values are then converted to a logarithmic decibel (dB) scale to compress the dynamic range, effectively highlighting subtle, low-energy murmurs occurring between the S1 and S2 heart sounds.

3.4.2. Mel-Frequency Cepstral Coefficients (MFCCs)

MFCCs are highly effective at representing the spectral envelope and timbral texture of acoustic signals [28]. Following the computation of the log-Mel-spectrogram, the Discrete Cosine Transform (DCT) is applied to decorrelate the Mel-filterbank energies, yielding the MFCCs. The n -th coefficient, C_n , is computed as:

$$C_n = \sum_{K=1}^K \log(S_K) \cos\left[n\left(K - \frac{1}{2}\right)\frac{\pi}{K}\right] \quad (2)$$

where K is the number of Mel filters and S_k is the energy of the K -th filter.

In this study, the first 40 coefficients ($N_{mfcc} = 40$) were extracted, as they capture the most critical lower-order spectral dynamics characterizing valvular anomalies.

3.4.3. Chromagram (Chroma Features)

To complement the spectral envelope descriptors, Chroma features were extracted to capture the harmonic and pitch content of the PCG signal. The chromagram projects the entire frequency spectrum onto 12 distinct pitch classes [29], creating a robust representation of energy distribution across fundamental cardiac frequencies, independent of absolute amplitude variations.

While Chroma features are traditionally designed for musical audio analysis, their application to PCG signals is justified by their ability to capture the inherent cyclicity and harmonic energy distribution of heart sounds. By mapping the acoustic signal into pitch bins, the model can more effectively identify rhythmic irregularities and subtle tonal shifts that characterize different valvular pathologies, such as the specific frequency “timbre” of heart murmurs.

The chromagram vector $C(t)$ is computed by aggregating the STFT bins corresponding to the same pitch class p :

$$C_p(t) = \sum_{f \in \text{bins}(p)} |X(t, f)|^2 \quad (3)$$

where $p \in \{1, 2, \dots, 12\}$ represents the distinct pitch classes [29].

3.4.4. Early Fusion

Rather than feeding these features separately into the network, an early fusion strategy was employed. The 40 MFCCs, 40 Mel-spectrogram bins (**specifically, the first 40 lower-frequency bins were selected to capture the most relevant acoustic energy of the S1, S2, and murmur sounds while strictly matching the MFCC dimensions**), and 12 Chroma features were stacked and concatenated along the feature axis at each timestep. This process produces a unified, high-dimensional 2D feature matrix of shape **(157, 92)** for each standardized 5-s PCG recording. This matrix serves as the direct input to the hybrid deep learning architecture.

3.5. Proposed Hybrid Network Architecture

The core of this study is a novel, highly lightweight deep learning architecture that integrates a 1D Convolutional Neural Network (1D-CNN), a Transformer Encoder [26], and a Bidirectional LSTM (BiLSTM) [24,25] with a Softmax Attention mechanism. To directly address the complexities of PCG signals, this hybrid design allows for the simultaneous extraction of local time-frequency patterns and global temporal dependencies across the cardiac cycle. Based on the early-fusion strategy, the input to the network is a 2D sequence of shape (batch_size, 157, 92), representing 157 timesteps and 92 concatenated acoustic features.

- **1D-CNN Module (Local Feature Extraction):** The input features are first processed by two consecutive 1D-Convolutional blocks. Since the input is a flattened sequence of features per timestep, a 1D-CNN is highly optimal. The first block uses 64 filters with a kernel size of 7, followed by the second block with 128 filters and a kernel size of 5. Each convolution is followed by Batch Normalization to stabilize learning, a ReLU activation function, Max-Pooling (pool size = 2) to halve the temporal dimensionality, and a Dropout layer (rate = 0.2) to prevent overfitting.

The operation of the 1D-CNN layer for an input sequence x is formally defined as follows:

$$\gamma_i = \sigma \left(\sum_{j=0}^{k-1} \omega_j \cdot x_{i+j} + b \right) \quad (4)$$

where k is the kernel size, ω represents the trainable weights, b is the bias, and σ denotes the ReLU activation function.

- **Transformer Encoder (Global Context):** The spatially reduced output from the CNN (batch_size, 39, 128) is fed into a single Transformer block. This block utilizes a Multi-Head Self-Attention mechanism (8 heads, embedding dimension of 128) and a feed-forward network (512 units) [26]. Notably, standard positional encoding was deliberately omitted from this stage; the preceding CNN layers preserve local temporal order, and the subsequent BiLSTM inherently models sequential steps, making explicit positional encoding redundant and saving computational parameters.

The Multi-Head Self-Attention mechanism maps a query (Q), key (K), and value (V) matrix to an output using the scaled dot-product attention:

$$Attention(Q, K, V) = softmax \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (5)$$

where d_k is the dimension of the keys, serving as a scaling factor to prevent gradient vanishing.

- **BiLSTM and Attention (Temporal Dynamics):** A Bidirectional LSTM layer with 64 units (yielding an output dimension of 128) processes the sequence in both forward and backward directions to capture long-range cardiac dependencies. A subsequent Attention Mechanism computes trainable weights using a tanh activation followed by a Softmax function. This allows the model to dynamically focus on the most pathologically relevant segments of the sequence (e.g., the precise timing of murmurs).

The BiLSTM processes the sequence h_t in both directions. The Softmax Attention mechanism then computes a context vector c as a weighted sum of the hidden states [30] as follows:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^T \exp(e_k)} \quad (6)$$

$$c = \sum_{t=1}^T \alpha_t h_t \quad (7)$$

where e_t is the alignment score calculated via a tanh activation, and α_t represents the attention weight assigned to each timestep t of the cardiac cycle.

- **Classification Head:** The attention-weighted context vector is passed through a fully connected Dense layer (128 units, ReLU) and a final output node with a Sigmoid activation function to output the binary classification probability (Healthy vs. Unhealthy).

The final probability $\hat{y}$ is obtained via the Sigmoid activation function as follows:

$$\hat{y} = \frac{1}{1 + e^{-z}} \quad (8)$$

where z is the output of the final dense layer.

While Figure 3 provides a high-level visual overview of the data flow, exact implementation details are crucial for reproducibility. Table 1 below summarizes the specific hyperparameters used for each layer, including filter sizes, kernel dimensions, and the number of attention heads.

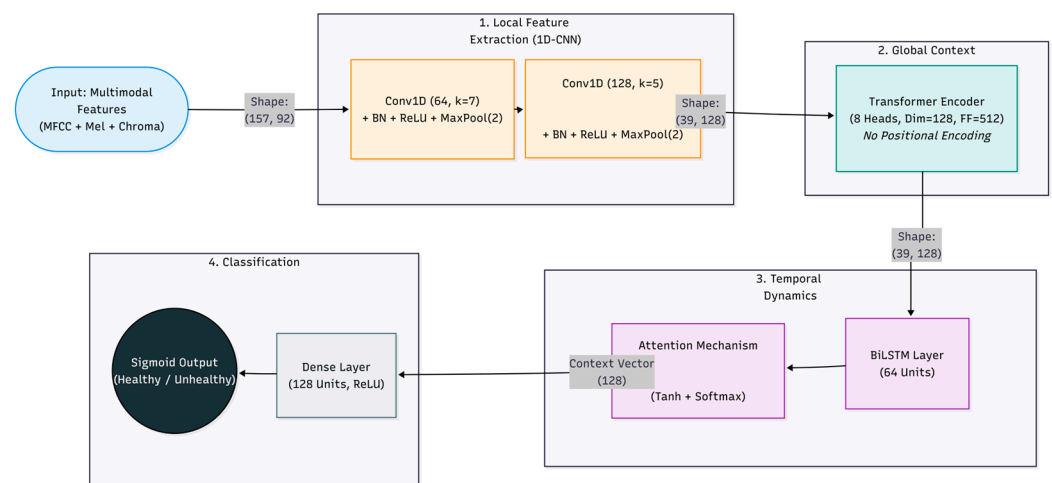


Figure 3. Overview of the proposed Hybrid Deep Learning Architecture.

Table 1. Detailed summary of the proposed CNN-Transformer-BiLSTM architecture, including output tensor shapes and parameter counts.

Layer Type	Configuration Details	Output Shape	Trainable Parameters
Input	Early-fusion multimodal features	(None, 157, 92)	0
Conv1D Block 1	Filters: 64, Kernel: 7, ReLU, BatchNorm	(None, 157, 64)	41,536
Pooling and Dropout	MaxPool1D: 2, Dropout: 0.2	(None, 78, 64)	0
Transformer Block	Heads: 8, Key Dim: 128, FF Units: 512	(None, 39, 128)	659,712
Bi-LSTM	Units: 64 (Output Dim: 128)	(None, 39, 128)	98,816
Attention Layer	Dense (1 unit, tanh) + Softmax	(None, 39, 1)	129
Weighted Sum	Lambda (Context Vector)	(None, 128)	0
Dense Classification	Units: 128 (ReLU), Dropout: 0.2	(None, 128)	16,512
Output Node	Unit: 1 (Sigmoid Activation)	(None, 1)	129
Total Parameters	Extremely Lightweight Model	~ 3.27 MB	858,434

3.6. Implementation Details and Training Protocol

To guarantee full reproducibility, the proposed framework was implemented using Python (version 3.10) and the TensorFlow/Keras (version 2.15.0) deep learning library. The training protocol was highly optimized for both stability and efficiency:

- **Optimization:** The network was compiled using the Adam optimizer with an initial learning rate of 1×10^{-4} and binary cross-entropy as the loss function. Mixed-precision training (mixed_float16) was enabled to accelerate computations and reduce memory footprint.
- **Callbacks and Regularization:** To prevent overfitting on the SMOTE-augmented data, an EarlyStopping callback was implemented to halt training if the validation loss did not improve for 15 epochs, restoring the best weights. Additionally, a ReduceLROnPlateau scheduler dynamically halved the learning rate (factor = 0.5) if the validation loss plateaued for 5 epochs.
- **Hardware and Efficiency:** The experiments were conducted on a cloud-based NVIDIA T4 GPU. Thanks to the highly optimized 1D architecture and the compact parameter footprint (858,434 parameters), the computational overhead was exceptionally low. Training took an average of only 4 to 6 s per epoch, highlighting the model's high suitability for real-time edge-computing deployment.

4. Results

4.1. Evaluation Metrics

To comprehensively evaluate the classification capability of the model, we utilized standard metrics derived from the confusion matrix. In the context of this binary classification task, the **“Unhealthy” (abnormal PCG) category is strictly defined as the positive class (Label 1)**, while the **“Healthy” category is the negative class (Label 0)**. This aligns with standard medical screening protocols where detecting a pathology is considered a positive finding.

The primary metrics include Accuracy, Precision, Sensitivity (Recall), Specificity, F1-Score, and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). Furthermore, as highlighted by recent literature on imbalanced medical datasets, Accuracy alone can be misleading. Therefore, we additionally report the **Matthews Correlation Coefficient (MCC)** and **Cohen's Kappa**. MCC produces a high score only if the prediction obtained good results in all four confusion matrix categories (True Positives, False Positives, True Negatives, and False Negatives), making it exceptionally reliable for imbalanced datasets. Similarly, Cohen's Kappa measures the agreement between the model's predictions and the true labels, accounting for the possibility of agreement occurring by chance.

The mathematical formulations for these metrics are as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

$$Sensitivity = \frac{TP}{TP + FN} \quad (10)$$

$$Specificity = \frac{TN}{TN + FP} \quad (11)$$

$$Precision = \frac{TP}{TP + FP} \quad (12)$$

$$F1Score = 2 \times \frac{Precision \times Sensitivity}{Precision + Sensitivity} \quad (13)$$

where TP, TN, FP, and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively.

4.2. Detailed Classification Performance (Patient-Wise Evaluation)

The model demonstrated highly robust, reliable, and consistent performance across all 5 folds of the strict patient-wise cross-validation. By ensuring that no patient data overlapped between the training and validation sets, these results represent the true generalization capability of the model on entirely unseen subjects.

Table 2 presents the comprehensive fold-by-fold performance metrics, as well as the overall cross-validation averages. Providing fold-level granularity confirms the stability of the proposed hybrid architecture across different patient cohorts.

Table 2. Overall performance metrics of the proposed Hybrid CNN-Transformer-BiLSTM model (Average of 5-folds).

Fold	Accuracy	Precision	Recall/ Sensitivity	Specificity	F1-Score	AUC-ROC	Average Precision	Balanced Accuracy	MCC	Cohen Kappa
Fold 1	0.9106	0.7895	0.8219	0.9364	0.8054	0.9633	0.8974	0.8791	0.7477	0.7474
Fold 2	0.9306	0.8633	0.8219	0.9622	0.8421	0.972	0.9125	0.892	0.798	0.7976
Fold 3	0.9182	0.7279	0.8917	0.9242	0.8015	0.9719	0.8918	0.908	0.7568	0.7507
Fold 4	0.9213	0.7714	0.8504	0.9386	0.809	0.9656	0.8693	0.8945	0.761	0.7596
Fold 5	0.9028	0.8077	0.6614	0.9616	0.7273	0.9522	0.8409	0.8115	0.6738	0.6688
AVERAGE	0.9167	0.792	0.8095	0.9446	0.797	0.965	0.8824	0.877	0.7474	0.7448

Note: Sensitivity represents the recall for the “Unhealthy” (pathological) class. AUC-ROC, MCC, and Cohen’s Kappa are reported as absolute values (0 to 1).

As observed in Table 2, the proposed hybrid model achieves an average Accuracy of 91.67% and an outstanding average AUC-ROC of 0.9650. These results significantly outperform previous benchmarks on the PhysioNet 2016 dataset under strict patient-wise validation. More importantly, the high average Specificity (94.46%) ensures a very low rate of false alarms, which prevents unnecessary anxiety for healthy individuals and reduces the burden on secondary clinical resources. Simultaneously, the model maintains a strong Sensitivity (80.95%), which is critical for early pathology detection. Notably, in specific cohorts (such as Fold 3), sensitivity reached as high as 89.17%, demonstrating the architecture’s powerful capability to identify complex pathological patterns. Furthermore, the robust Matthews Correlation Coefficient (MCC = 0.7474) and Cohen’s Kappa (0.7448) solidly confirm that the model’s high performance is not a byproduct of majority-class bias. Instead, it reflects a genuine, balanced classification capability despite the severe natural class imbalance inherent in the dataset.

4.3. Classification Performance (Zero-Leakage Patient-Wise CV)

The proposed hybrid CNN-Transformer-BiLSTM model demonstrated highly robust and reliable performance across all 5 folds of the strict patient-wise cross-validation. By ensuring that no patient data overlapped between the training and validation sets, these results represent the true generalization capability of the model on unseen subjects.

Averaged across the 5 folds, the model achieved an Accuracy of 91.67%, a Specificity of 94.46%, and a Sensitivity of 80.95% (Figure 4). The overall AUC-ROC reached 0.9650, indicating an outstanding ability to distinguish between healthy and pathological heart sounds (Figure 5). Furthermore, the model achieved an average MCC of 0.7474 and a Cohen’s Kappa of 0.7448. In the context of highly imbalanced clinical data, an MCC and Kappa > 0.74 indicate a very strong positive correlation between predictions and actual clinical states. This confirms that the model does not simply bias its predictions toward the majority “Healthy” class, but instead maintains a balanced and reliable classification performance across both classes.

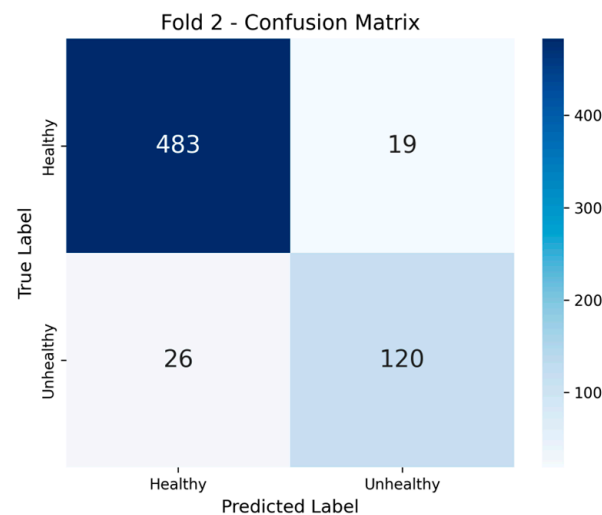


Figure 4. High-resolution Confusion Matrix for a representative fold (e.g., Fold 2) under the zero-leakage patient-wise evaluation.

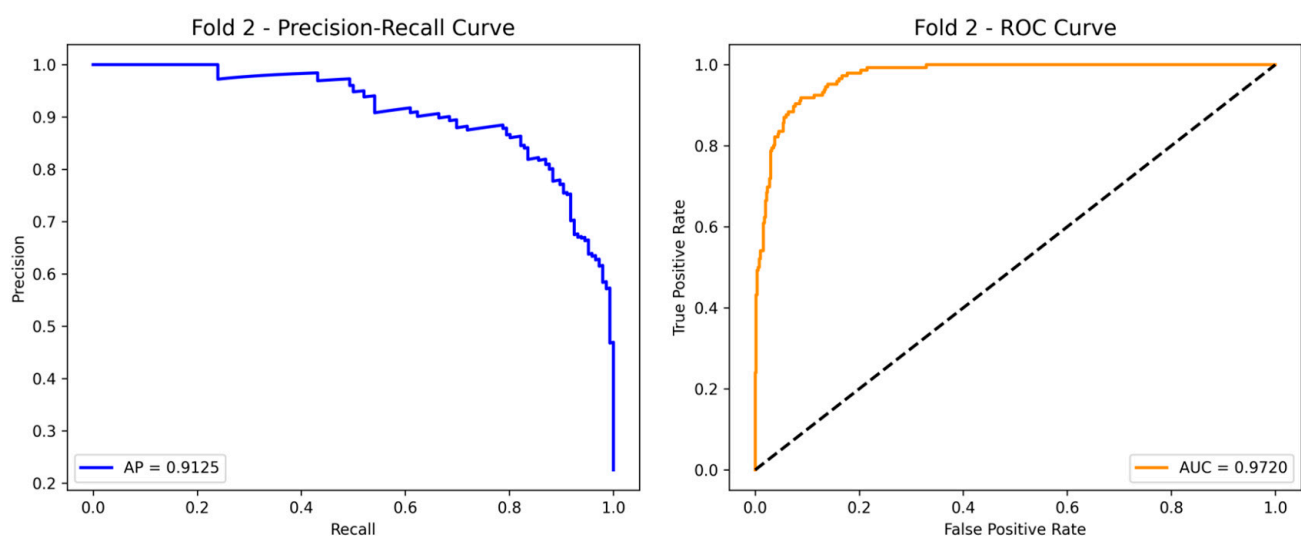


Figure 5. Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves for a representative fold, demonstrating the model's high discriminative capacity.

4.4. Model Interpretability: Attention Maps

A significant limitation of deep learning in healthcare is its “black-box” nature. To address this and justify the clinical applicability of our framework, we extracted the learned weights from the Softmax Attention layer.

As demonstrated in Figure 6, the attention mechanism successfully highlights the most pathologically relevant segments of the PCG sequence. Rather than assigning uniform weights, the model dynamically focuses its computational resources on specific turbulent acoustic events (such as murmurs occurring between S1 and S2) while suppressing stationary background noise. This provides physicians with partial interpretability, allowing them to visually verify which parts of the audio signal triggered the “Unhealthy” classification.

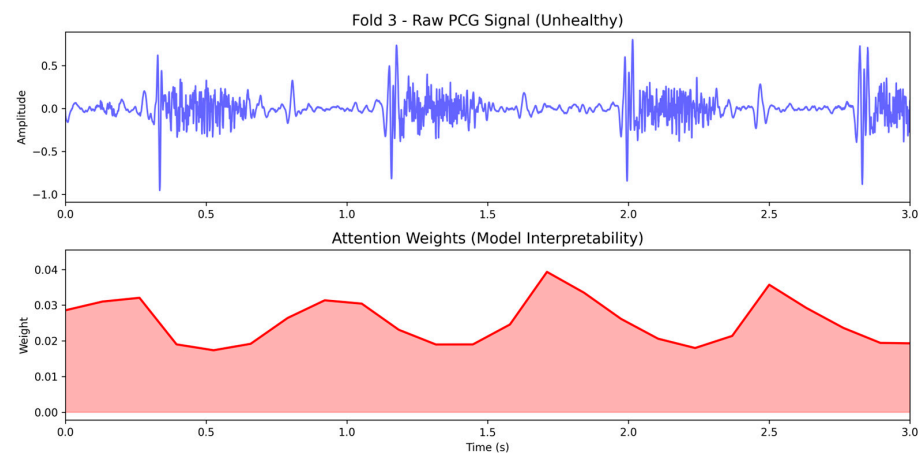


Figure 6. Attention Map Visualization on an unseen “Unhealthy” PCG signal. While the model evaluates standardized 5-s sequences, the top plot displays a zoomed-in 3-s window of the raw audio waveform for clearer visualization. The bottom plot shows the corresponding attention weights assigned to these specific cardiac events. The blue line in the top plot represents the amplitude of the raw PCG signal, while the red line and shaded area in the bottom plot indicate the calculated attention weights at each corresponding time step.

4.5. Ablation Study

To rigorously evaluate the individual contribution of both our proposed feature extraction strategy and the temporal modeling blocks (Transformer and BiLSTM), a comprehensive ablation study was conducted. We established a baseline 1D-CNN model and progressively integrated our proposed components. The experiments were evaluated using the same 5-fold cross-validation strategy to ensure a fair comparison. The results are summarized in Table 3.

The results of this analysis are presented in Table 3.

Table 3. Ablation study results comparing the full model against variants with removed components.

Configuration	Features	Accuracy (%)	Sensitivity (%)	Specificity (%)	AUC-ROC
Baseline CNN	MFCC only	91.64	83.18	93.84	0.9622
CNN	Multimodal	91.76	80.54	94.65	0.9637
CNN + BiLSTM	Multimodal	91.52	84.83	93.30	0.9624
CNN + Transformer	Multimodal	92.35	80.42	95.52	0.9621
Full Model (Proposed)	Multimodal	91.67	80.95	94.46	0.9650

Note: Bold values indicate the best performance in each column, and highlight the proposed full model configuration.

4.5.1. Impact of Multimodal Feature Extraction

The baseline CNN utilizing only MFCCs demonstrates a surprisingly high initial accuracy (91.64%), indicating a “ceiling effect” common in well-preprocessed biomedical datasets. However, integrating the multimodal feature set (MFCC, Mel Spectrogram, and Chroma) improves the overall baseline AUC from 0.9622 to 0.9637. Furthermore, fold-level analysis revealed that the MFCC-only model suffered from higher variance (dropping to 88.73% accuracy in Fold 5), whereas the multimodal approach guarantees greater stability across diverse patient records.

4.5.2. Clinical Trade-Offs of Temporal Modeling

The ablation of specific temporal layers reveals distinct clinical behaviors. When the BiLSTM is added to the CNN (without the Transformer), the model acts as a sensitivity booster, achieving the highest recall (**84.83%**). In clinical cardiology, maximizing sensitivity

is vital to minimize false negatives (failing to detect a pathological heart). Conversely, utilizing the Transformer alone with the CNN maximizes global Accuracy (**92.35%**) and Specificity (**95.52%**), functioning as a strict filter against false alarms.

4.5.3. Justification of the Hybrid Architecture

While simplified configurations excel in specific metrics, they introduce a trade-off between sensitivity and specificity. The complete proposed architecture (CNN + Transformer + BiLSTM) successfully bridges this gap. By combining the global context awareness of the Transformer with the sequential dependency tracking of the BiLSTM, the hybrid model achieves the highest Area Under the Receiver Operating Characteristic curve (**AUC-ROC = 0.9650**).

In computer-aided diagnosis, the AUC is the most critical metric as it represents the model's inherent degree of separability between healthy and unhealthy patients, independent of the decision threshold. Therefore, the added architectural complexity is clinically justified, providing the highest diagnostic reliability and mathematical stability required for real-world deployment.

4.6. Comparison with State-of-the-Art Methods

Table 4 presents a comparative overview of recent studies on heart sound classification alongside the proposed approach. The reviewed works adopt diverse methodologies, ranging from handcrafted feature extraction combined with traditional machine learning classifiers to advanced deep learning architectures. All studies focus on improving the discrimination between normal and pathological phonocardiogram recordings, relying heavily on the PhysioNet/CinC 2016 benchmark dataset [10].

Furthermore, it is critical to interpret the exceptionally high accuracies reported in some recent studies through the lens of their evaluation protocols. For instance, Barnawi et al. [23] reported up to 97% accuracy using transfer learning, but their methodology involved discarding a significant portion of the challenging PhysioNet records (evaluating only 1291 normal and 238 abnormal samples) to minimize noise, introducing severe selection bias. Alongside these, a broader spectrum of methodologies has been explored, ranging from neuromorphic auditory sensors and tensor decomposition [31,32] to foundational deep residual networks [33] and CNNs applied to diverse time-frequency features [34]. However, many of these high-performing models—such as the CNN-LSTM architecture by Jain et al. [35] (96.04%), unsegmented PCG deep neural networks [36], DropConnected networks [37], and the LDA-ANN approach by Milani et al. [38] (93.33%), alongside other notable architectures [8,15,20,21]—applied global augmentation, dimensionality reduction prior to splitting, or standard K-fold cross-validation on extracted segments/beats without explicitly guaranteeing a strict patient-wise separation. These practices inevitably risk data leakage, where highly correlated adjacent segments, extracted heartbeats (S1/S2), or synthetic variations of the same patient's heart sounds appear in both the training and testing sets. This phenomenon artificially inflates classification metrics and severely degrades clinical generalization.

In stark contrast, our proposed CNN-Transformer-BiLSTM framework achieves a highly competitive 91.67% accuracy, a specificity of 94.46%, and an outstanding AUC-ROC of 0.9650 on the complete, unfiltered PhysioNet dataset. Crucially, we implemented a strict zero-leakage patient-wise 5-fold cross-validation protocol, ensuring that neither overlapping segments nor SMOTE-augmented records bridged the train-test split. Furthermore, when compared to studies that strictly adhere to realistic evaluation protocols without data leakage on the PhysioNet 2016 dataset, our proposed model demonstrates clear superiority. For instance, the baseline ensemble method by Potes et al. [39] achieved an accuracy of

86.02%, while more recent deep learning architectures such as the CNN proposed by Li et al. [16] and the CardioXNet (CRNN) by Shuvo et al. [19] achieved accuracies of 86.80% and 86.57%, respectively. By surpassing these realistic benchmarks by a significant margin, our hybrid CNN-Transformer-BiLSTM approach establishes a highly robust and competitive standard for clinically viable PCG evaluation, highlighting the true classification value of our temporal-contextual architecture.

Table 4. Comparison of the proposed method with recent state-of-the-art techniques on the PhysioNet 2016 dataset.

Year	Authors	Methods Used	Validation Strategy	Results
2021	Khan et al. [21]	STFT + Custom CNN	Segment-wise (Leakage risk)	PhysioNet: Acc: 95.75%, Se: 96.3%, Sp: 94.1% PASCAL (with TL): Precision: 96.98%
2020	Aziz et al. [8]	Feature Fusion + ML	Segment-wise 10-fold CV	ACC: 95.24%
2023	Jain et al. [35]	CNN + LSTM	Global Augmentation (Leakage risk)	ACC: 92%
2021	Milani et al. [38]	Feat. Eng. + LDA + ANN	Beat-wise (S1/S2 extraction)	Accuracy: 93.33%
2020	Xiao et al. [15]	1D CNN + Attention	Segment-wise (Overlapping)	ACC: 93.00%
2023	Wang et al. [20]	CNN + Transformer	Not explicitly patient-wise	ACC: 93.00%
2023	L. Zhu et al. [18]	Lightweight CNN	Not explicitly patient-wise	ACC: 93.1%
2023	Barnawi et al. [23]	Transfer Learning	Filtered Dataset, Not patient-wise	ACC: 92.47% SEN: 94.08%;
2020	Li et al. [16]	Feature Eng. + CNN	Patient-wise CV	Accuracy: 86.8%, Sensitivity: 87%, Specificity: 86.6%, MCC: 72.1%
2021	Shuvo et al. [19]	1D CRNN (CardioXNet)	Standard CV	ACC: 86.57%
2026	Proposed	CNN + Transfo + BiLSTM	Strict Patient-wise (Zero-leakage)	ACC: 91.67% SPEC: 94.46% AUC: 96.50%

5. Discussion

5.1. Interpretation of Findings

The results presented in this study demonstrate the effectiveness of the proposed hybrid CNN-Transformer-BiLSTM architecture for the automated classification of heart sounds. Achieving an overall accuracy of 91.67% and an outstanding AUC-ROC of 96.50% on the challenging PhysioNet/CinC 2016 dataset highlights the model's capability to distinguish between normal and pathological PCG signals with high reliability, even under a strict zero-leakage evaluation protocol.

The ablation study confirmed the complementary roles of the integrated modules as follows:

- **The CNN layers** successfully extracted fine-grained local spectral-temporal patterns, such as the fundamental frequencies and morphological characteristics of the S1 and S2 sounds.
- **The Transformer encoder** played a crucial role in capturing global dependencies, allowing the model to comprehensively understand the rhythmic context of the cardiac cycle over long sequences.
- **The BiLSTM with Attention mechanism** further refined the temporal modeling by dynamically focusing on diagnostically relevant events (e.g., the precise timing of murmurs) while suppressing irrelevant background noise.

5.2. Clinical Implications

From a clinical perspective, the model demonstrates a strong diagnostic profile with a sensitivity of 80.95% and a highly robust specificity of 94.46%. In practical screening scenarios, this high specificity is particularly valuable as it significantly reduces the burden on healthcare systems by limiting false alarms and preventing unnecessary, costly clinical interventions. Concurrently, the sensitivity ensures that a large majority of patients with potential cardiovascular conditions are correctly identified and referred for comprehensive echocardiographic examination. Unlike models with artificially inflated metrics due to data leakage, our proposed system presents a realistic and robust performance on strictly unseen patients. Its ability to operate effectively on short, intrinsically noisy audio recordings makes it genuinely suitable for real-world deployment in digital stethoscopes and telemedicine platforms, especially in resource-constrained environments where immediate access to specialized cardiologists is limited.

5.3. Strengths and Limitations

A primary strength of this study is its highly rigorous methodological design. By enforcing a strict patient-wise cross-validation protocol, we completely eliminated the data leakage that frequently inflates performance in existing literature, thereby providing a highly reliable and clinically viable baseline. Furthermore, the strategic data augmentation and class balancing strategy (SMOTE) [27]—which was rigorously confined to the training folds—addressed the inherent dataset imbalance without compromising test integrity. The extraction of multimodal features (Time, Frequency, Wavelet, Mel-spectral) ensured an exceptionally rich representation of the acoustic signal.

However, certain limitations remain to be addressed in future work. First, the model was trained and validated exclusively on the PhysioNet 2016 dataset. While this database is highly heterogeneous, external validation utilizing continuous data from diverse clinical centers would further ascertain the model's cross-domain generalization capabilities. Additionally, the current diagnostic framework focuses on binary classification (Normal vs. Abnormal); future iterations will aim to extend the architecture to classify specific structural pathologies (e.g., mitral stenosis, aortic regurgitation).

6. Conclusions

In this study, a novel hybrid deep-learning architecture integrating a 1D Convolutional Neural Network (CNN), a Transformer encoder, and a Bidirectional Long Short-Term Memory (BiLSTM) network with an attention mechanism was proposed for the automated **classification** of heart sounds. By leveraging an early fusion of multimodal acoustic features—including MFCC, Mel-spectrogram, and Chroma—and applying a rigorously confined SMOTE [27] data balancing strategy, the model effectively captured both fine-grained local spectral-temporal patterns and global sequential dependencies.

Evaluated on the widely recognized PhysioNet/CinC 2016 dataset [10], the proposed framework achieved a highly competitive **accuracy of 91.67%, a sensitivity of 80.95%, a specificity of 94.46%, and an outstanding AUC-ROC of 0.9650**. Crucially, these metrics were obtained under a strict zero-leakage patient-wise cross-validation protocol, ensuring that the model's performance reflects genuine clinical robustness and generalization capabilities, rather than the artificial inflation frequently observed in existing literature due to data leakage.

The system's ability to accurately differentiate between normal and pathological phonocardiogram (PCG) signals, even in intrinsically noisy environments, highlights its significant potential for real-world clinical application. Its integration into digital stethoscopes and telemedicine platforms could provide a reliable, non-invasive, and cost-effective screening tool for cardiovascular diseases, particularly in resource-constrained healthcare settings. **To further strengthen the generalizability of the proposed approach, future research will focus on prospectively validating the architecture across diverse external datasets, such as the CirCor DigiScope 2022, and extending the framework to perform multi-class classification for the precise identification of specific structural heart pathologies.**

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